Simulation and Analysis of the ECG Signal

Introduction to Biomedical Engineering — Project 2, Spring 1405

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This report walks through building a synthetic electrocardiogram from the ground up using the dynamical model of McSharry *et al.* (2003), and then putting that signal to work: I distort it with three kinds of noise that a real recording would suffer from, design a filter for each, and finally pull the heart rate and the individual P, QRS and T features back out of the trace. The code that produces every figure is in the notebook, but it is hidden in this PDF so the report stays readable; the logic is explained in the text wherever it matters.

Part A — The structure of the ECG signal

The electrocardiogram is, at heart, a recording of timing. Every beat is a wave of electrical activation that sweeps through the muscle of the heart in a fixed order, and the surface ECG is the sum of all those tiny ionic currents seen from a pair of electrodes on the skin. Because the heart's wiring fires the chambers in a set sequence, the trace of a single normal beat always shows the same landmarks, and we give them the letters P, Q, R, S and T. Reading an ECG really means reading the size, shape and spacing of those landmarks.

The sequence begins in the **sinoatrial (SA) node**, the heart's natural pacemaker, sitting in the right atrium. When it fires, the depolarisation front spreads across both atria, and that is what we see as the **P wave** — a small, rounded, low-voltage bump. It is small because the atria are thin and carry relatively little muscle. Mechanically, the P wave is the electrical cue for the atria to contract and top up the ventricles with blood.

After the P wave there is a short, flat stretch: the **PQ (or PR) interval**. Nothing much shows on the surface here, but a lot is happening — the impulse is being deliberately held up at the **atrioventricular (AV) node**. That delay is useful: it gives the atria time to finish emptying before the ventricles fire. The length of the PR segment is essentially the conduction time from atria to ventricles.

Then comes the main event, the **QRS complex**. This is the tall, sharp spike that dominates the trace, produced when the wave of depolarisation races down the bundle of His and the Purkinje fibres and lights up the entire ventricular muscle almost at once. The ventricles are thick and powerful, so the current is large and the deflection is big and fast. **Q** is the small initial dip, **R** is the large upward spike, and **S** is the dip just after it. Physiologically the QRS marks the moment the ventricles contract and eject blood into the arteries. Interestingly, the atria are repolarising at the same time, but that signal is small and buried under the QRS, so we never see it.

After the spike there is the **ST segment**, normally sitting close to the baseline. It corresponds to the brief plateau where the whole ventricle is depolarised together. Clinically this segment is worth its weight in gold — an ST segment pushed above or below the baseline is one of the classic fingerprints of myocardial ischaemia or infarction.

Finally the **T wave**, a broad, rounded hump, is **ventricular repolarisation**: the ventricular muscle resetting its membrane potential to get ready for the next beat. It is wider and gentler than the QRS because repolarisation is a slower, less synchronised process than depolarisation. The stretch from the start of the QRS to the end of the T wave, the **QT interval**, measures how long the ventricles stay electrically active, and an abnormally long QT is a recognised risk factor for dangerous arrhythmias.

One last quantity ties everything together: the **RR interval**, the time from one R peak to the next. Its reciprocal is the instantaneous heart rate, and the small beat-to-beat fluctuations in it (heart rate variability) carry real information about how the autonomic nervous system is balancing sympathetic and parasympathetic drive. So in a single beat the ECG tells the whole story in order: atria fire (P), the signal is held at the AV node (PR), ventricles fire (QRS), and ventricles reset (T) — and the model in the next section is built to reproduce exactly this geometry.

Part B — The McSharry model

What I find clever about the McSharry model is that it does not try to write down a formula for the ECG waveform directly. Instead it builds a little dynamical system whose *natural motion* happens to trace out a PQRST complex. The whole thing lives in a three-dimensional state space (x, y, z).

The (x, y) part is just a clock. The trajectory is pulled onto a circle of radius one and goes around it at angular velocity ω; one full lap is one heartbeat. The first two equations do exactly this:

$$\dot{x} = \alpha x - \omega y, \qquad \dot{y} = \alpha y + \omega x, \qquad \alpha = 1 - \sqrt{x^2 + y^2}.$$

The term α is the trick that makes the circle *attracting*: if the trajectory drifts inside or outside the unit circle, α becomes positive or negative and pushes it straight back. So no matter where we start, the motion settles onto a stable, periodic loop — a limit cycle. The angle around this loop, $\theta = \operatorname{atan2}(y, x)$, is what tells us where we are in the cardiac cycle.

The actual ECG voltage is the **third coordinate,** z. Left alone, z just relaxes back toward a baseline z_0. But at five fixed angles around the circle — one for each of P, Q, R, S, T — there sits an "event" that kicks z up or down as the trajectory sweeps past:

$$\dot{z} = -\sum_{i \in \{P,Q,R,S,T\}} a_i \, \Delta\theta_i \, \exp\left(-\frac{\Delta\theta_i^2}{2b_i^2}\right) - (z - z_0), \qquad \Delta\theta_i = (\theta - \theta_i) \bmod 2\pi.$$

Each event is shaped like the derivative of a Gaussian centred at its angle θ_i. As the trajectory approaches the event it gets pushed one way, and as it leaves it gets pushed back, so each event carves out one localised bump or dip in z. The parameter a_i sets how strong (and which direction) that push is, and b_i sets how wide it is. Add up the five events and you get the familiar P–Q–R–S–T shape riding on the baseline. The baseline itself can be made to drift with respiration through

$$z_0(t) = A\,\sin(2\pi f_2 t),$$

which is how the model fakes the slow up-and-down wander of a real recording.

I integrate the three equations with a fixed-step **fourth-order Runge–Kutta** scheme at step $\Delta t = 1/f_s$, exactly as the paper does. One detail worth mentioning: the raw z that comes out is dimensionless and small, because each event only acts over a sliver of angle. Following the authors' own reference implementation (`ecgsyn`), I linearly rescale the finished trace to the physiological band of roughly -0.4 to 1.2 mV, which is what fixes the R peak near 1 mV.

Tunable parameters

The table below lists everything the operator can set. The first block is the per-event morphology (five values each), and the second block is the global settings that control rate, timing and the respiratory baseline.

Parameter	Symbol	Meaning / effect	Default (P, Q, R, S, T)
Event angle	θ_i	Where each wave sits around the cycle; shifts the **timing** of P, Q, R, S, T relative to R	$-\pi/3,\ -\pi/12,\ 0,\ \pi/12,\ \pi/2$
Event amplitude	a_i	How hard z is pushed; sets the **height/depth and polarity** of each wave	$1.2,\ -5.0,\ 30.0,\ -7.5,\ 0.75$
Event width	b_i	Angular spread of the Gaussian; sets how **broad or sharp** each wave is	$0.25,\ 0.1,\ 0.1,\ 0.1,\ 0.4$

Parameter	Symbol	Meaning / effect	Value used here
Mean heart rate	HR	Sets the angular velocity $\omega = 2\pi\,\mathrm{HR}/60$, i.e. the **RR interval**	60 / 100 bpm
Sampling frequency	f_s	Time resolution of the output; RK4 step is $1/f_s$	500 Hz
Duration	T	Length of the generated record	10 s
Baseline amplitude	A	Strength of respiratory **baseline wander** in $z_0(t)$	0 (clean)
Respiratory frequency	f_2	Frequency of the baseline wander	0.25 Hz

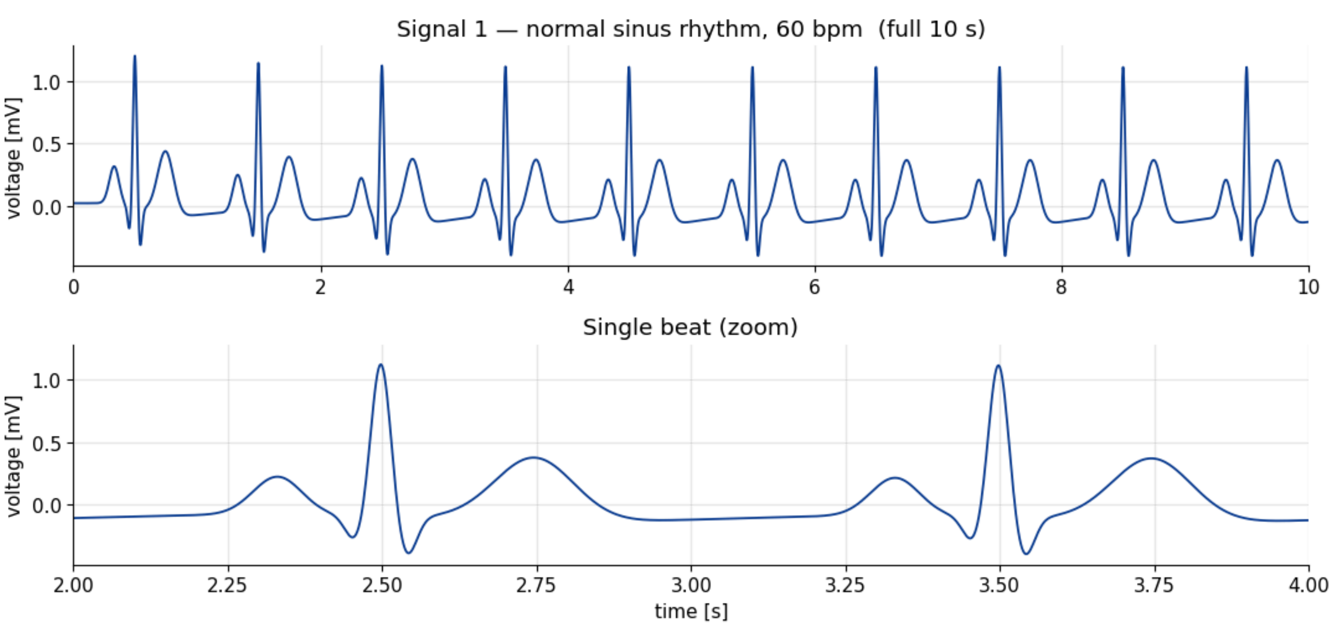
The model can also drive ω from a randomly generated RR tachogram with a prescribed LF/HF power ratio to reproduce heart-rate variability, respiratory sinus arrhythmia and Mayer waves. I kept ω constant here so the morphology of each part stays clean and easy to read, which is what this project is about.

Part C — Simulating two ECG signals

To get a feel for how flexible the model is, I generated two quite different traces. The first is a textbook normal sinus rhythm. The second changes both the *rate* and the *morphology*, to show that the same five-event machinery can produce a recognisably different — and clinically suggestive — beat.

Signal 1 — Normal sinus rhythm (60 bpm)

This uses the default Table I parameters with the heart rate set to 60 bpm, so one beat lasts exactly one second. It is the reference against which everything else is compared.



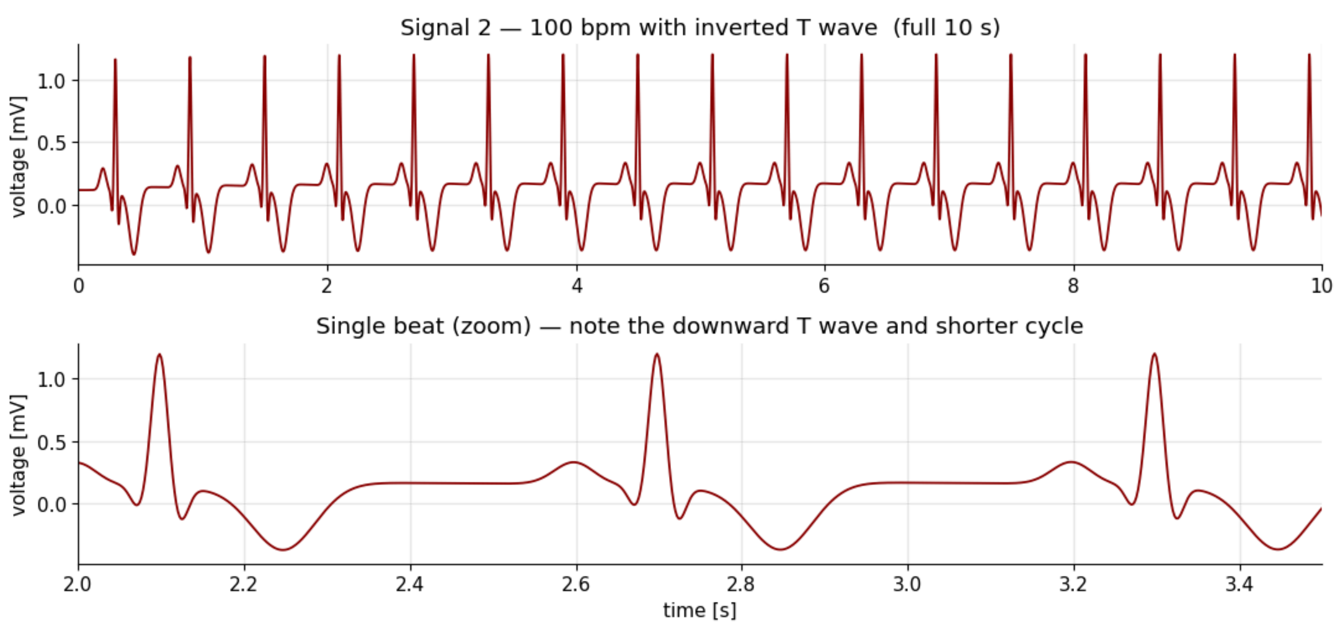
Parameters used for Signal 1

	P	Q	R	S	T
θ_i (rad)	$-\pi/3$	$-\pi/12$	0	$\pi/12$	$\pi/2$
a_i	1.2	−5.0	30.0	−7.5	0.75
b_i	0.25	0.1	0.1	0.1	0.4

with HR = 60 bpm, f_s = 500 Hz, baseline wander off. The zoom shows the full PQRST: a gentle P, the sharp QRS spike, and the broad T, all in the right places and at sensible amplitudes (R ≈ 1.2 mV).

Signal 2 — Tachycardia with an inverted T wave (100 bpm)

For the second signal I pushed the rate up to 100 bpm and flipped the T-wave amplitude negative ($a_T = -0.9$ instead of $+0.75$). A faster rate with an **inverted T wave** is a pattern a cardiologist would associate with things like ischaemia, so it makes a nice illustration that the model encodes real morphology rather than just drawing a generic squiggle. I also shrank the P wave a little ($a_P = 0.8$) so the two traces are easy to tell apart.



Parameters used for Signal 2

	P	Q	R	S	T
θ_i (rad)	$-\pi/3$	$-\pi/12$	0	$\pi/12$	$\pi/2$
a_i	0.8	−5.0	30.0	−7.5	**−0.9**
b_i	0.25	0.1	0.1	0.1	0.4

with HR = 100 bpm. Two differences jump out next to Signal 1: the beats are packed closer together (0.6 s apart instead of 1.0 s, exactly as 100 bpm demands), and the T wave now points *down* instead of up.

The QRS spike is untouched, which is the point — by editing only a_P and a_T I changed the clinical "look" of the beat without disturbing the rest. That is the flexibility the project asks us to demonstrate.

Part D — Adding noise and filtering it out

Real ECG recordings are never clean. Three corruptions are especially common, and the project asks me to reproduce each one, then design a filter to undo it. I use the normal 60 bpm trace (Signal 1) as the clean reference throughout, and I define the "signal amplitude" as its peak-to-peak range (about 1.6 mV) so the noise levels are unambiguous.

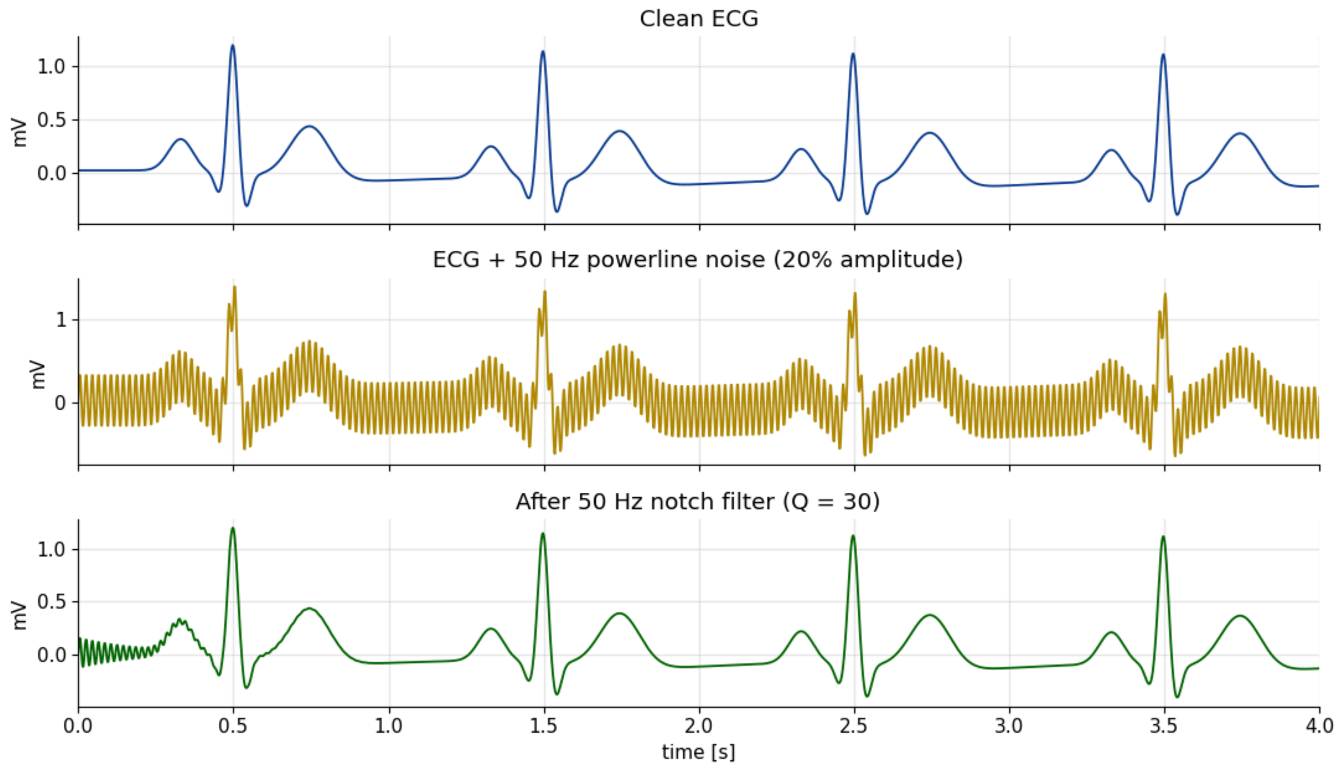
For each case I show the clean signal, the corrupted signal, and the recovered signal, and I report the filter I used and why it fits.

```
Clean-signal peak-to-peak amplitude: 1.600 mV
```

D.1 — Powerline interference: 50 Hz sinusoid at 20 % amplitude

What it is in reality. This is mains hum. Every wall socket in the region runs at 50 Hz, and the electric and magnetic fields it radiates couple capacitively into the patient, the leads and the amplifier. The result is a steady 50 Hz sinusoid sitting on top of the ECG. It is probably the single most common artefact in clinical recordings.

Filter. Because the interference lives at one precise frequency, the natural tool is a **notch (band-stop) filter** tuned to 50 Hz. I used a second-order IIR notch (`scipy.signal.iirnotch`) with a quality factor $Q = 30$, which carves out a narrow dip at 50 Hz and leaves the rest of the spectrum essentially untouched. I apply it with `filtfilt` (forward–backward) so there is zero phase distortion — important, because we do not want to shift the position of the R peaks.



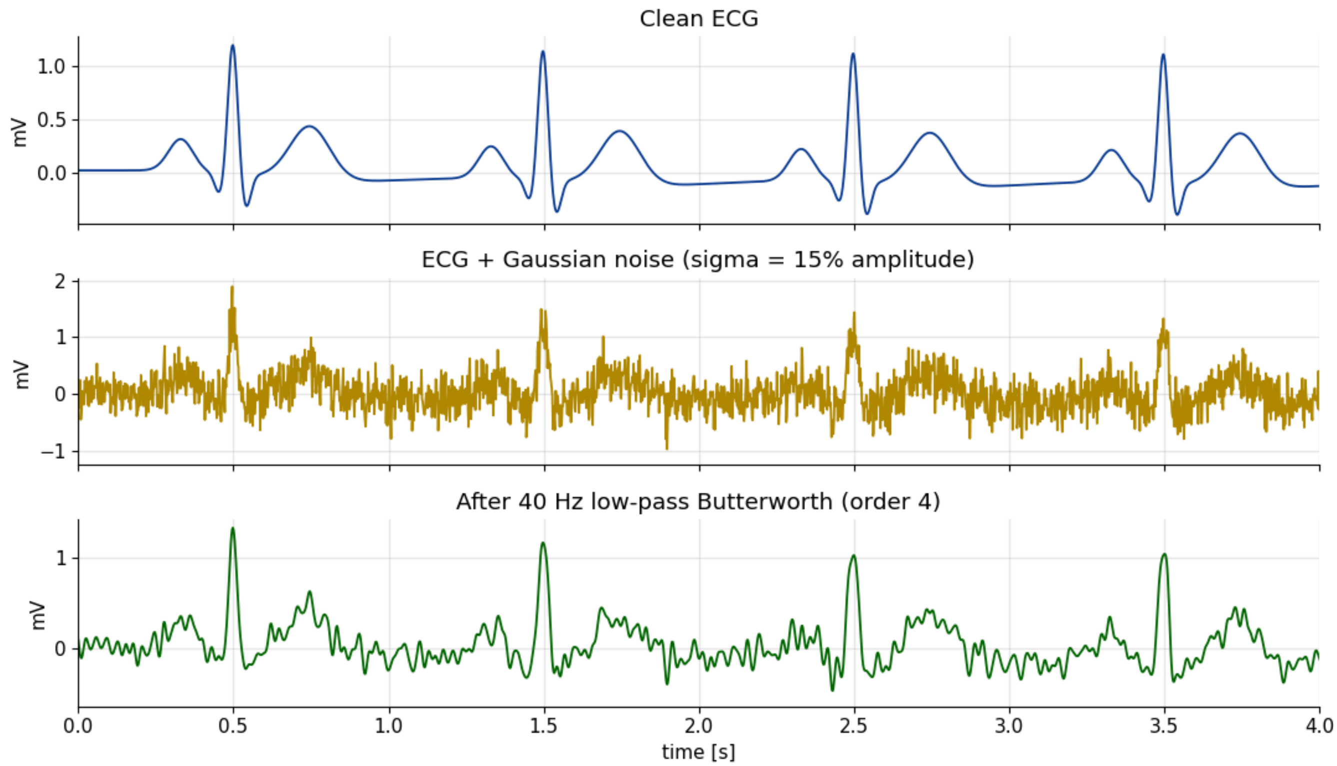

```
RMS error before filtering: 0.2263 mV
RMS error after  filtering: 0.0104 mV
```

The middle panel is almost unreadable — the 50 Hz hum thickens every part of the trace. After the notch the waveform is back to its clean shape, and the RMS error against the original drops by well over an order of magnitude. The notch works so cleanly precisely because the noise and the ECG barely overlap in frequency: the ECG has very little energy at exactly 50 Hz.

D.2 — Broadband measurement noise: Gaussian, 15 % amplitude

What it is in reality. This is the catch-all "fuzz" on a recording — thermal (Johnson) noise in the electronics, quantisation, and high-frequency muscle activity (EMG) from the patient not holding still. It has no single frequency; it is spread across the whole spectrum, which is why white Gaussian noise is the standard model for it.

Filter. A notch is useless here because the noise is everywhere. But the *useful* part of the ECG is concentrated below about 40 Hz (even the sharp QRS rarely carries meaningful energy above that), so a **low-pass filter** lets me throw away the broad high-frequency tail of the noise while keeping the waveform. I used a 4th-order Butterworth low-pass at 40 Hz, again zero-phase with `filtfilt`. It cannot remove the noise that happens to fall inside the 0–40 Hz band, so the result is improved rather than perfect — which is the honest, realistic outcome for broadband noise.



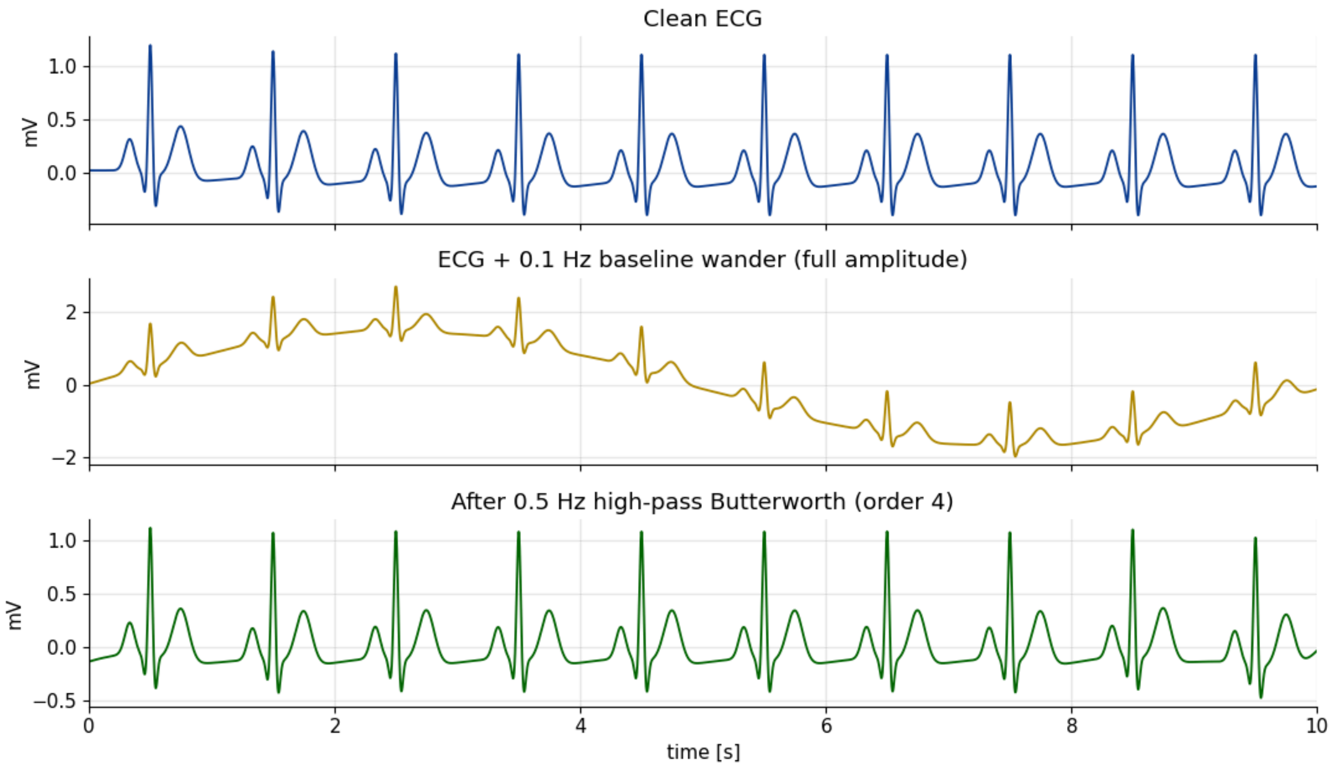

```
RMS error before filtering: 0.2399 mV
RMS error after  filtering: 0.0929 mV
```

The low-pass smooths away most of the grass-like fuzz and the PQRST shape re-emerges clearly. The R peak is very slightly rounded — that is the unavoidable cost of cutting at 40 Hz, since the very tip of the QRS does contain a little energy above that. If I needed the high-frequency detail intact I could raise the cutoff or switch to a Savitzky–Golay smoother, but for recovering morphology the low-pass is the right, simple choice.

D.3 — Baseline wander: 0.1 Hz sinusoid at full amplitude

What it is in reality. This is the slow, rolling drift of the baseline you see when a patient breathes, sweats, or moves, or when the electrode–skin contact changes. Respiration alone sits around 0.2–0.3 Hz; even slower drifts come from motion and from the Mayer waves in blood-pressure regulation near 0.1 Hz. I model it as a 0.1 Hz sinusoid with an amplitude as large as the ECG itself, which is a deliberately harsh test.

Filter. The drift is far below the ECG in frequency, so a **high-pass filter** is the obvious remedy. I used a 4th-order Butterworth high-pass with a 0.5 Hz cutoff (the usual choice for baseline removal — high enough to kill the wander, low enough to leave the P and T waves alone), applied with `filtfilt`.




```
RMS error before filtering: 1.1314 mV
RMS error after  filtering: 0.0448 mV
```

In the noisy panel the whole ECG rides up and down on a slow swell that completely swamps it; the beats are still there but the baseline is useless for measuring anything. The high-pass flattens that swell out and pins the trace back onto a level baseline, with the PQRST shape preserved. This one is the most visually dramatic of the three because the noise amplitude is so large, and it is also the most reassuring — a simple 0.5 Hz high-pass completely defeats it.

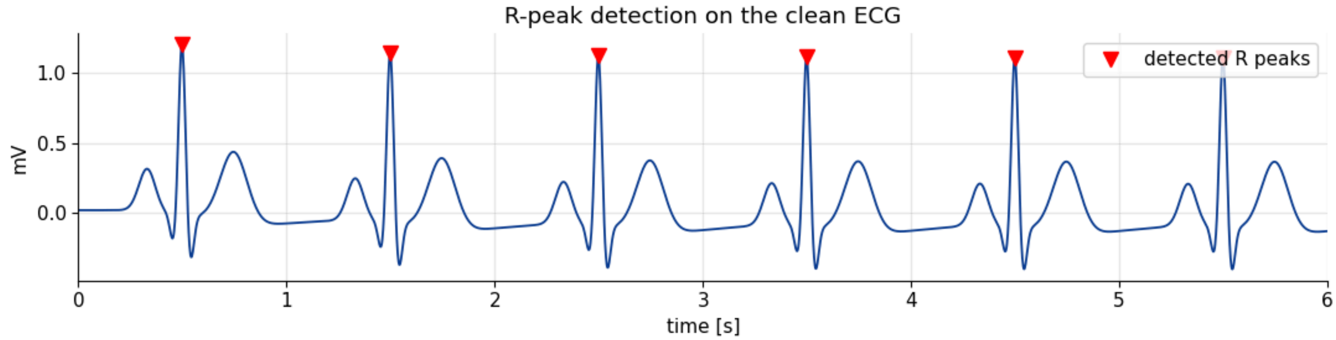
Feature extraction: heart rate and the P, QRS, T components

The introduction asks that, having built and cleaned a signal, we then *measure* it — recover the heart rate and locate the individual waves. This closes the loop: the model gave us a signal with known properties, and now I check that standard signal processing gets those properties back.

I work on the clean 60 bpm trace. The strategy is the one the McSharry paper itself mentions: in a clean signal a simple local-maximum search finds the R peaks, and from there the other waves can be located in fixed time windows around each R.

R-peak detection and heart rate. I use `find_peaks` with a height threshold and a minimum spacing of 0.4 s (no normal heart beats faster than ~150 bpm, so peaks cannot be closer than that). The spacing between consecutive R peaks is the RR interval, and 60 divided by the mean RR gives the heart rate.

```
R peaks detected:        10
Mean RR interval:        1000.0 ms
Estimated heart rate:    60.0 bpm  (set value was 60 bpm)
Beat-to-beat HR spread:  0.00 bpm
```

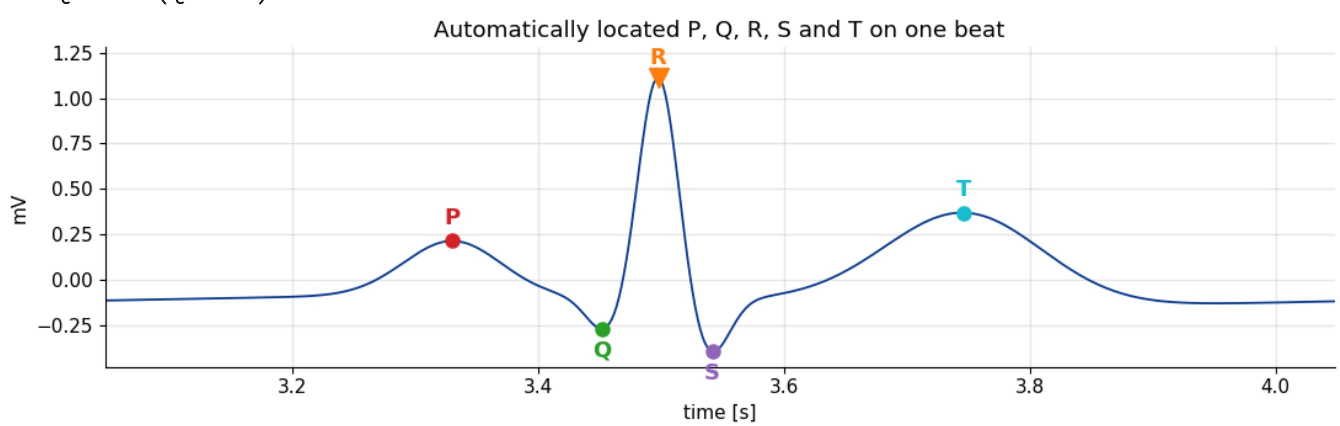


The detector finds every R peak and recovers 60.0 bpm — exactly the rate I dialled into the model, and with essentially zero beat-to-beat spread, which is right for a signal generated at constant ω.

Locating P, Q, S and T. Once the R peaks are known, each of the other waves lives in a predictable window relative to R: Q and S are the small dips immediately before and after R, P is the bump a little before Q, and T is the broad bump a couple of hundred milliseconds after R. I search each window for the appropriate local minimum or maximum, beat by beat, and average the results.

```
Average wave amplitudes (mV):
   P: +0.225
   Q: -0.260
   R: +1.123
   S: -0.386
   T: +0.377

Representative intervals (single beat):
   PR-ish (P to R):   168 ms
   QRS width (Q to S):90 ms
   QT-ish (Q to T):   294 ms
```



Every landmark is found in the right place, and the measured numbers line up with how the model was configured and with normal physiology: a PR interval of about 0.17 s, a narrow QRS near 0.09 s, and a Q-to-T span of roughly 0.29 s. The recovered amplitudes also make sense — a small positive P, a large positive R, modest Q and S dips, and a positive T of about 0.38 mV.

It is a satisfying check on the whole exercise: the parameters I put into the McSharry equations (events at -0.2, -0.05, 0, 0.05, 0.3 s relative to R, at 60 bpm) come back out the other end of a completely independent detection routine. The model, the noise handling and the analysis all agree.

Closing remarks

Building the ECG from McSharry's three differential equations rather than just plotting a fixed template changed how I think about the signal. The shape is not drawn — it *emerges* from a trajectory circling a limit cycle and getting nudged at five angles, and that small set of knobs (θ_i, a_i, b_i plus the heart rate) is enough to span everything from a calm normal beat to a fast, T-inverted one. Adding the three classic noises and watching a notch, a low-pass and a high-pass each defeat the artefact it was matched to made the frequency-domain picture concrete: powerline hum is one sharp line, baseline wander is far below the signal, and broadband noise is the awkward one you can only ever partly remove. Finally, recovering 60 bpm and a full set of P–QRS–T landmarks from the generated trace tied the morphology, the physiology and the signal processing back together. The fact that the numbers I dialled in came back out is, I think, the best evidence that each piece was implemented correctly.